

Part One - Single Choice

- Event A and event B are independent, and the probability that at least one of A, B occurs is $\frac{8}{9}$. It is known that the probability that A occurs and B does not occur is equal to the probability that A does not occur and B occurs, then the probability of A, i.e., $P(A)$, equals ().
 (A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) $\frac{4}{9}$ (D) $\frac{5}{9}$
- If $F(x) = \frac{2}{2+x^2}$ is the cumulative distribution function of random variable X, then the value range of X is ().
 (A) $(-\infty, \infty)$ (B) $(0, \infty)$ (C) $(-\infty, 0)$ (D) $(0, 1)$
- Random variable X follows an exponential distribution with parameter 1, i.e., $X \sim \text{Exp}(1)$, a is a constant greater than zero, then $P(X \leq a+1 | X > a)$ equals ().
 (A) $1-e^{-1}$ (B) e^{-1} (C) $1-e^{-a}$ (D) e^{-a}
- D is a flat area enclosed by the curve $y = \frac{1}{x}$ and straight lines $y=0, x=1, x=e^2$. The two-dimensional random variable (X, Y) follows a uniform distribution on D, then the value of the marginal density function of (X, Y) for X at $x=2$, i.e., $f_X(2) = ()$.
 (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{e}$ (D) $\frac{1}{e^2}$
- Random variables X and Y are independent, $X \sim N(0, 1)$, $Y \sim N(1, 1)$, then which of the following statements is correct ().
 (A) $P(X-Y \leq 0) = 0.5$ (B) $P(X-Y \leq 1) = 0.5$ (C) $P(X+Y \leq 0) = 0.5$ (D) $P(X+Y \leq 1) = 0.5$

Part Two - Blank Filling

- Let A, B, C be three mutually independent events, with $P(A) = P(B) = \frac{1}{2}$, $P(C) = \frac{1}{3}$. Then the probability that at most one of the events A, B, C happens is _____.
- Distribute randomly 6 chopsticks from 3 different pairs to three persons. Then the probability of at least one person getting two chopsticks from the same pair is _____.
- Take two balls without replacement from 3 red balls, 2 black balls and 1 white ball. Then the probability that both balls are red is _____.
- Let X be a Poisson random variable with parameter $\lambda = \ln 2$ and let $F(x)$ be its cumulative distribution function. Then $F(1) = \underline{\hspace{2cm}}$.
- Toss a fair coin 10 times. Let A be the event of getting exactly 2 heads in total, and let B be the event of getting all tails in the first 5 tosses. Then $P(B|A) = \underline{\hspace{2cm}}$.

6. Suppose that X is a random variable with cumulative distribution function $F(x) = ae^{bx}/(1+2e^x)$. Then $a = \underline{\hspace{2cm}}$, $b = \underline{\hspace{2cm}}$.
7. In an exam that requires 60 to pass, suppose that the score of student A has the normal distribution $N(65, a^2)$ and the score of student B has the normal distribution $N(60+a, 10^2)$. If the probability of A passing the exam is at least as high as B doing so, then the range for a is $\underline{\hspace{2cm}}$.
8. Suppose that X has the uniform distribution over $(0, 1)$, and given X , Y has the uniform distribution over $(0, X)$. Consider the events $A = \{Y \geq \frac{1}{3}\}$ and $B = \{X \leq \frac{2}{3}\}$. Then $P(B|A) = \underline{\hspace{2cm}}$.
9. Two independent random variables X and Y satisfy that $X, Y \in \{0, 1\}$ and $P(X=0, Y=1) = \frac{1}{4}$, $P(X=1, Y=0) = \frac{1}{6}$. Then $P(X=0, Y=0) = \underline{\hspace{2cm}}$.
10. Let X and Y be independent and be exponential random variables with parameters 1 and 2, respectively. Let $Z = \max(X, Y)$. Then for $z \geq 0$, the density function for Z is $f_z(z) = \underline{\hspace{2cm}}$.

Part Three — Question Answering

1. Let the sample space be Ω . Now consider two events A and B , with the probability of occurrence being respectively 0.4 and 0.7.
- (1) Are A and B disjoint? Prove your answer;
 - (2) If $A \cup B = \Omega$, are A and B independent? Prove your answer;
 - (3) Now consider a geometric model of probability. Let $\Omega = [0, 1]$ be the unit interval in $\mathbb{R}$, and suppose $A = [0, 0.4]$, $B = [a, b]$, $0 \leq a \leq b \leq 1$. If A and B independent, find the values of a and b .
2. An insurance company divides the insured into three categories: "cautious", "average" and "rash". Statistics show that the probabilities of accidents for these three types of people within a year are 0.05, 0.15 and 0.3 respectively. If "cautious" insured people account for 20%, "average" people account for 50%, and "rash" people 30%.
- (1) What ~~it~~ is the probability that an insured person will be involved in an accident within a year?
 - (2) If it is known that an insured person has been involved in an accident, find the probability that he belongs to the "cautious" category.
3. Suppose that the random variable X follows normal distribution, $X \sim N(5, 25)$.
- (1) Compute $P(3 < X < 7)$;

(2) Find x such that $P(X > x) = 0.39$;

(3) Find the value range and the density function of $Y = e^X$.

[Note: Standard normal distribution table $\Phi(0.26) = 0.6$; $\Phi(0.28) = 0.61$; $\Phi(0.30) = 0.62$;
 $\Phi(0.33) = 0.63$; $\Phi(0.36) = 0.64$; $\Phi(0.39) = 0.65$; $\Phi(0.4) = 0.66$; $\Phi(0.44) = 0.67$]

4. Let the joint frequency function of the two-dimensional random variable (X, Y) be

$X \backslash Y$	0	1	2
0	0.06	0.15	α
1	β	0.35	0.21

(1) What conditions do you need to meet for the constant α and β ?

(2) If X and Y are independent, please calculate the values of α and β ;

(3) If X and Y are independent, please find the marginal frequency functions of X and Y , respectively.

5. Let X and Y have the joint density function:

$$f(x, y) = \begin{cases} 1, & |x| < y, 0 < y < 1 \\ 0, & \text{others} \end{cases}$$

(1) Find the marginal densities $f_X(x)$ and $f_Y(y)$.

(2) Are X and Y independent?

(3) Find the conditional densities function $f_{X|Y}(x|y)$ and $f_{Y|X}(y|x)$.

6. Suppose that the random variables X and Y are independent, where $X \sim U(0, 1)$, $Y \sim U(0, 2)$.

(1) Find the probability of $P(X \leq Y)$.

(2) Find the distribution function of $X + Y$.